

QUALITY ASSURANCE & TESTING DOCUMENT

MedAI

QA & Testing Document

Comprehensive test strategy, test cases, acceptance criteria, and validation procedures for the MedAI mobile application and hospital web portal.

Document: QATD v1.0 **Scope:** Mobile App + Web Portal **Test Types:** Functional, Integration, E2E, Performance, Security

SECTION 1 — TEST STRATEGY

Testing Strategy

4 Test levels	48 Test cases defined	5 Critical user flows	100% P0 coverage required
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Test Pyramid

Unit Tests Model serialization, AI service parsing logic, state machine transitions, utility functions. Tool: flutter_test	Integration Tests Firebase Auth flow, Firestore read/write, Storage upload, real API responses. Tool: flutter_test + Firebase Emulator Suite	End-to-End Tests Full user flows on real device/emulator: registration → AI chat → booking → appointments view. Tool: flutter drive / Maestro
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Test Entry / Exit Criteria

☒ Entry Criteria - Feature branch merged to main - flutter pub get succeeds with no errors - flutter analyze returns no errors (warnings acceptable) - Firebase project configured (test environment) - Test device / emulator available	☒ Exit Criteria - All P0 test cases pass - No crash on happy path flows - AI chat completes end-to-end on real API - Appointment appears in Firestore after booking - Performance: app launch < 3s cold start
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Risk Matrix

Risk	Probability	Impact	Mitigation
AI API downtime / rate limit	Medium	High	Fallback verdict; retry logic; user-facing error message
Firestore security rules too restrictive	Medium	High	Test rules with Firebase Emulator before deploying
Composite index missing for queries	High	Medium	Deploy all indexes via firestore.indexes.json pre-launch
File picker permissions denied	Medium	Low	Handle PermissionDeniedException; show settings deep-link
Slot cache expired / empty	Medium	Medium	Graceful fallback message; re-seed mock data on demand

SECTION 2 — FUNCTIONAL TEST CASES

Authentication Test Cases

ID	Test Case	Steps	Expected	Priority
TC-AUTH-01	Successful registration	Fill all fields correctly → tap Register	User created in Firebase Auth + Firestore /users; routed to Home	P0
TC-AUTH-02	Duplicate email registration	Register with existing email	Error banner: This email is already registered	P0
TC-AUTH-03	Invalid email format	Enter notanemail in email field	Inline validation error; form not submitted	P1
TC-AUTH-04	Successful sign-in	Enter valid credentials → Sign In	Routed to Home; AuthProvider.isAuthenticated = true	P0
TC-AUTH-05	Wrong password	Enter valid email + wrong password	Error: Incorrect password	P0
TC-AUTH-06	Forgot password flow	Tap Forgot password → enter email → Submit	Success snackbar; Firebase sends reset email	P1
TC-AUTH-07	Session persistence	Sign in → close app → reopen	User remains authenticated; Home shown (not Sign In)	P0
TC-AUTH-08	Sign out	Profile → Sign Out → Confirm	Firebase signOut called; routed to Sign In screen	P0

AI Chat Test Cases

ID	Test Case	Expected Result	Priority
TC-CHAT-01	Chat initializes with greeting	Opening AI Chat tab shows hardcoded greeting message with AI avatar	P0
TC-CHAT-02	Symptom collection — full flow	After 3–6 AI exchanges, AI detects SYMPTOMS_COMPLETE; transitions to verdict state	P0
TC-CHAT-03	Verdict shown correctly	AI message shows specialty + urgency + explanation + disclaimer	P0
TC-CHAT-04	Booking prompt displayed	Card with 'Yes, book now' and 'Maybe later' buttons appears	P0
TC-CHAT-05	Decline booking flow	Tapping 'Maybe later' → hardcoded closing message; chat saved to Firestore	P0
TC-CHAT-06	Full booking flow	Accept → collect prefs → show slots → select slot → confirmation card shown	P0
TC-CHAT-07	Appointment created in Firestore	After slot selection, document exists in /appointments with status=confirmed	P0
TC-CHAT-08	Document attachment — PDF	File picker opens; PDF selected; uploaded to Storage; document card shown; AI acknowledges	P1
TC-CHAT-09	Reset chat	AppBar refresh → confirm dialog → new greeting shown; previous messages cleared	P1
TC-CHAT-10	AI API failure graceful fallback	On network error, error snackbar; chat state unchanged; retry possible	P1

SECTION 3 — INTEGRATION, PERFORMANCE & ACCEPTANCE

Appointments Test Cases

ID	Test Case	Expected Result	Priority
TC-APPT-01	Appointment appears after booking	Booking via AI chat → appointment visible in Upcoming tab without refresh	P0
TC-APPT-02	Cancel appointment	Cancel → dialog → confirm → status=cancelled → moves to Past tab	P0
TC-APPT-03	Empty state — no appointments	New user sees illustration + 'No appointments yet' + CTA button	P1
TC-APPT-04	Real-time update	Appointment status updated in Firestore → reflected in UI within 2s without user action	P1

Profile Test Cases

ID	Test Case	Expected Result	Priority
TC-PROF-01	Avatar upload	Camera/gallery picker → image uploaded to Storage → avatarUrl updated in Firestore → shown in profile	P1
TC-PROF-02	Edit full name	Edit dialog → save → Firestore updated → profile shows new name	P1
TC-PROF-03	Change password	Enter current + new password → Firebase updatePassword → success snackbar	P1

Performance Test Targets

Metric	Target	Critical Threshold	Measurement Method
Cold app launch to Home	< 2.5s	< 4s	Flutter DevTools Timeline
Sign-in response time	< 1.5s	< 3s	Stopwatch from tap to navigation
AI first response (symptom)	< 4s	< 8s	Timestamp delta in ChatProvider
Appointment list load (Firestore stream)	< 1s	< 2s	isLoading flag timing
File upload (PDF, 2MB)	< 5s	< 15s	Storage putFile duration
App frame rate	60 fps steady	> 45 fps avg	Flutter DevTools Performance overlay
APK size	< 30MB	< 50MB	flutter build apk --analyze-size

Acceptance Criteria (Definition of Done)

Must Pass (P0)	Should Pass (P1)
- User can register and immediately use the app - AI chat completes full symptom → verdict → booking flow - Created appointment visible in Firestore and Appointments tab - Cancel appointment works; status updated in Firestore - Session persists across app restarts - No crash on any P0 test case - Onboarding shown once; never again after completion	- PDF/document attachment uploads and AI acknowledges - Reset chat clears conversation and starts fresh - Profile edits persist to Firestore - Avatar upload works via camera and gallery - Error states shown gracefully (no blank screens) - Forgot password email sent successfully - Home screen shows correct next appointment

Device Test Matrix

Device / OS	Priority	Test Scope
Android Emulator (API 34, x86_64)	P0	All test cases — primary development target
Android Physical (Samsung Galaxy, API 31)	P1	P0 test cases + camera/gallery picker
iOS Simulator (iPhone 15, iOS 17)	P1	All test cases (requires Mac + Xcode)
Small screen (360×640, Android)	P2	Layout verification, no overflow